# Guide for the execution of the project in the context of the course "Software Technology"

## Before you begin:

### Team Selection

As straight forward as it may seem, choosing the right team is probably the most important step that will determine whether your project will be successful. From the very beginning ALL team members, ideally 3-4, should be willing to work, barring the unexpected, on one of the deliverables. There should be an easy understanding between the members on how and when everyone should do their part and of course so that any changes can be made and checked in time. Finally, from the beginning each member should identify his/her strengths and weaknesses so that he/she can undertake the tasks that can be completed within time.

### Selection of Topic

For the choice of the topic, originality or the reinvention of the wheel is not as important as usability. That is why the first task of the project is to find the Use Cases of the project. For example, many groups when we did the assignment (Spring 2020-2021), chose to capitalize on the coronavirus situation and created some kind of Reservation System which while not new could find infinite applications from Restaurants to the university library. Another good example of TL work is CEID's CoCoTr@ce system as well as some kind of Video Game, since it is a great example of the Software Development process.

### Working Procedure

The work is divided into 6 Deliverables (5 partial and a 6th final) each with 2-3 weeks difference from the previous one. After each delivery within a reasonable time (in theory) you will receive feedback either from the lecturers or from one of the course partners, the so-called "Angels". This feedback will be a short text that will tell you each time what you have done right in the delivery and if there is any addition or correction needed. And with this feedback you will proceed with the assignment.

### Preparation and Tools

As far as the preparation for the Project is concerned, all that is needed is the creation of a common Repository in a Version Control System (e.g. Git) and a first thought about the programming language in which the Project will be implemented. Instructors recommend Object Oriented Programming Languages (e.g. Java, C++, Python) so you will need to find which language works best for you. For creating diagrams a good solution that many people have used is Draw.io and Lucidchart and I would suggest that you familiarise yourself a little with their interface before starting Project to make it easier to make the requested diagrams. For creating Mockups I personally used Mockflow which has a free plan enough for the needs of the Project. Finally for writing the technical texts I think the best tools are Google

Docs for writing in simple word type texts and Overleaf if you want to work in LaTex, since both offer multi-user collaboration features.

# 1st Deliverable

### Team Plan

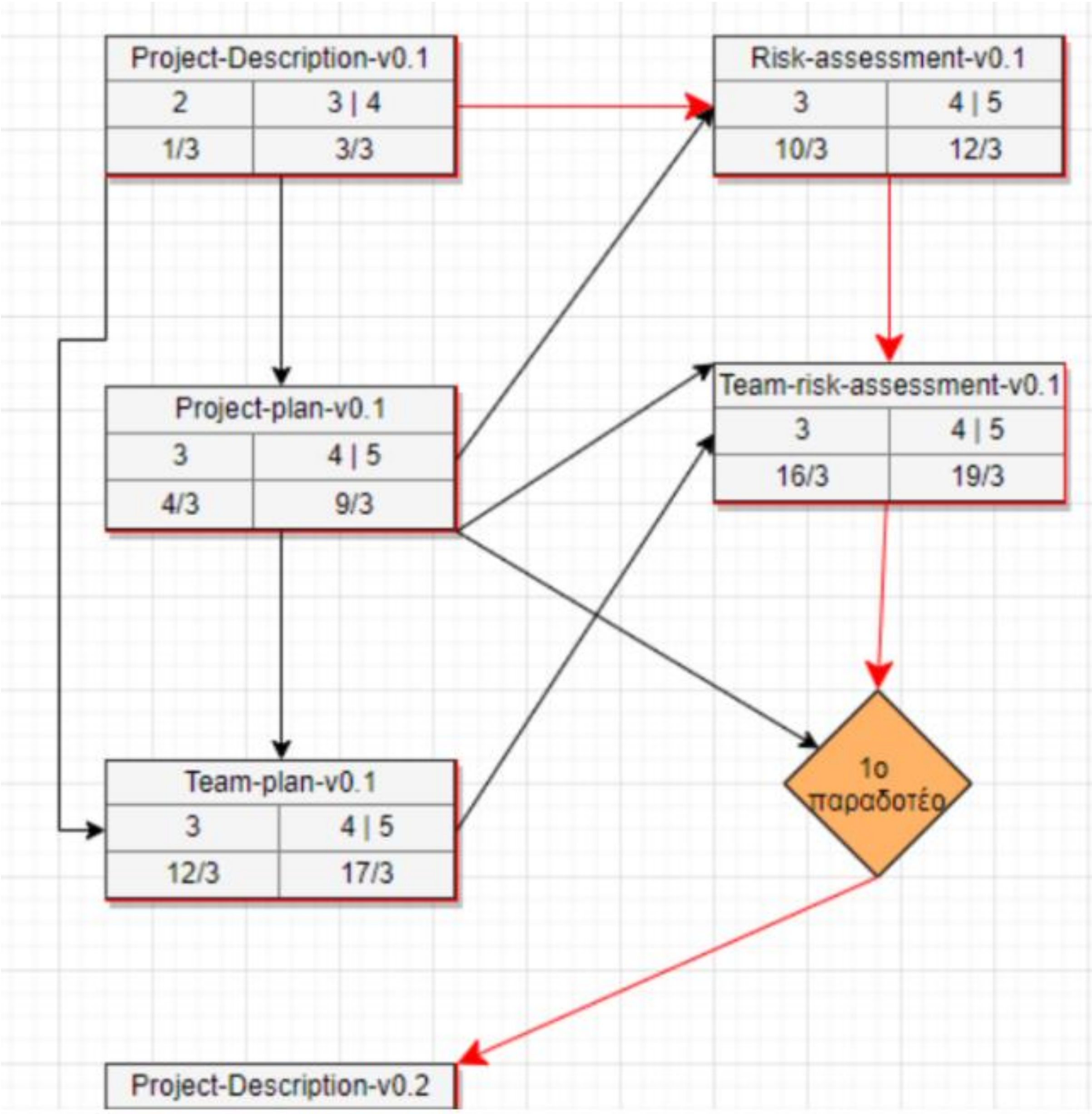
In this technical document you should first indicate the composition of your team. A simple table with Names, Academic credentials and contact details is sufficient.

Then you need to provide the Gantt and Pert charts for scheduling the Project. The Gantt Chart is simply a basic timeline for when you plan to implement each task of the project. An example can be seen below.


**Gantt Diagram**

The columns of the chart will include each of the 6 deliverables, divided into its individual requests. On the right side of the diagram are tiles representing the duration of each task and the order in which they will be implemented. At the end of each task a yellow diamond is placed, which represents the achievement of a Milestone (In our case the delivery of the task part).

Continuing with the Pert Chart, it is essentially the same content as the Gantt chart, but in the form of a flowchart.




**Pert Diagram**

As you can see in the image, the pieces of each deliverable are inserted in a tile that includes the name of the document, the order in which the tasks will be implemented (top left), the time of implementation (top right) and the start and end dates of the individual project. As far as the arrows connecting the tiles are concerned, these are the dependencies between the tasks. Tasks linked by black members can be implemented in parallel, while the dependency with a red arrow indicates that one task must finish before its dependent task can start.

Finally, the Team Plan asks you to describe the method you will use. As an example, you are given the SCRUM method. This obviously does not have to be implemented, although it would be good to follow it. In this section you simply select and describe a method.

**Project Description**

In this document you will include a brief description of your idea as you would present it to its end user. In the example of the Reservation System we gave at the beginning, you should present its functionality (what our system does) and how it would be useful for both the user and the customer (streamlining the seat reservation process).

Then you should also present some Mock-Ups on what you imagine your program/application will look like. These Mock-Ups can be either Wireframe or Complete. This mostly depends on how confident you are in your design, and how much you want to work on the designs.

**Project Plan**

In this text you will have to reimplement Gantt and Pert diagrams. This time, however, instead of the individual members of the deliverable, the diagrams will contain the tasks that you think will be done during the development process. You can for example include concept conception, team building, tool selection, mockups, testing, etc. The method by which you will implement the diagrams is the same as the Team Plan.

You should then create a simple table that includes "Assignment of a project to Human Resources.". In this table you will create a column with all the tasks from your Gantt chart and then assign one or more members of your team to do this task. Obviously, like other technical documents, this assignment can change from version to version.

Finally, you should make a theoretical cost estimate including things like staff salaries, any hosting you may need, internet connection etc.

**Risk Assessment**

In the Risk Assessment technical document, you will create a table in which you will include possible risks that you may encounter during the implementation of your project and possible solutions. These risks can be many and varied and depend mainly on the nature of your idea. Returning to the Reservation System example, one risk may be the lack of potential partners or even the existence of high competition (as we have seen during the development of the project).

**Team Risk Assessment**

In this text the idea is similar to the Risk Assessment text, but this time you will analyze risks that can be caused by your team, for example a conflict between two team members or the departure of a member during the project.